Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/28/2006
PI Name:	Issacci, Farrokh Ph.D.		
Project Title:	Two-Phase Flow in Multi-Channels - Liquid Holdup and Capillary Flow		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Farrokh.Issacci@honeywell.com	**Fax:**	FY 310-512-3432
PI Organization Type:	INDUSTRY	**Phone:**	310-512-1883
Organization Name:	Honeywell International		
PI Address 1:	MS 36-1-93193		
PI Address 2:	2525 West 190th Street		
PI Web Page:			
City:	Torrance	**State:**	CA
Zip Code:	90504	**Congressional District:**	35
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	06/12/2000	**End Date:**	12/01/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	The study will use a low-gravity environment to single out inertial and capillary forces in two-phase flow through rectangular channels with different cross-sectional dimensions, from sub-millimeter to centimeters. These configurations are common in many flow systems of space and terrestrial applications. The low-gravity environment provided by the NASA KC-135 Low Gravity Aircraft will allow for investigation of two-phase flow in larger channels (macro-channels) where inertia may be an important effect. The laboratory experiments in small channels (micro-channels) will provide information on two-phase flow where inertia may be negligible. The micro-channel experiments will provide data that can be used to verify the low-gravity findings as well as theoretical modeling. The study will combine theoretical studies with experimentation. The technical approach will include: identification of the needs for model development (based on applications and design issues), development of a test plan for normal and low-gravity experimentation, performance of experiments, comparison of theoretical results with the experimental data, and modifications/enhancement of existing		

Task Description:	models based on experimental data. This is a ground-based investigation of the fundamental aspects of two-phase flow behavior in a multi-channel flow manifold. Some aspects of this investigation are the liquid holdup phenomena in the stagnation points, as well as the capillary effects on the flow characteristics and surface tension driven flow. This study will provide basic information required in space applications for environment control and life support systems (ECLSS) such as fuel cells, bioreactors, thermal management systems, phase separators, etc. The benefit of this program to NASA and its mission is development of modeling tools to develop space systems. The results of this study will be the building blocks of the design tools necessary for the development of critical technologies for space power and life support systems, as well as terrestrial applications such as more efficient fuel cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)